NRF TAPEOUT #2 — KLAYOUT LAYER & DESIGN GUIDELINE

Standardized Positive-Logic Layout Specification for Analog & Device Designers

DOC: NRF-TO2-KLAYOUT-REV1
SUBSTRATE: P-Type Bulk Si

- **100% Positive Logic ("Draw What You Want"):** If you want a material, doped region, or opening on the wafer, simply draw a polygon on that layer. The Photomask Integration Team will convert your GDSII into the final fabrication photomasks (handling all negative resist inversions, mask combinations, and process-level Boolean operations).
- **All 8 Layers Required in Final GDSII:** Even if your device never uses a layer (e.g., `SI_ETCH`, `VIA1`, or `M2`), your exported `.gds` file must still include all 8 layer records defined in `NRF_Tapeout2.lyp` (enable *"Save empty layers"* when exporting, and store your device GDSII file as `Tapeout2_<DeviceName>.gds`).

1. MANDATORY 8-LAYER KLAYOUT STACK

KLayout Layer	GDS Layer	What Drawing This Layer Gives You on the Final Wafer	Physical Result on Wafer (After Photomask Conversion)
`1. SI_ETCH`	`1/0`	**Etched Silicon Trench (~200 nm)**	Creates a ~200 nm deep etched trench directly in the bulk silicon (used primarily for optical alignment markers; most devices leave this empty).
`2. NPLUS`	`2/0`	**Heavily Doped n+ Silicon Region**	Produces a heavily doped **n+** phosphorus silicon region on the final wafer wherever drawn.
`3. NWELL`	`3/0`	**Moderately Doped n-Type (N-Well) Region**	Produces a moderately doped **n-type** phosphorus silicon region on the final wafer wherever drawn.
`4. THINOX`	`4/0`	**Thin Dry Oxide (~40 nm SiO_2)**	Etches away the thick (~400 nm) thermally grown oxide and grows a thin **~40 nm dry oxide** in its place wherever drawn.
`5. CONT`	`5/0`	**Oxide Opening to Silicon**	Etches an opening through the thermally grown silicon dioxide down to the silicon surface wherever drawn.
`6. M1`	`6/0`	**Metal 1 Aluminum (~60 nm Al)**	Deposits and patterns the first layer of **~60 nm sputtered Aluminum** on the wafer surface wherever drawn.
`7. VIA1`	`7/0`	**Opening in ~1 µm PVD SiO_2**	Etches an opening through the ~1 µm sputtered PVD SiO_2 dielectric layer wherever drawn.
`8. M2`	`8/0`	**Metal 2 Aluminum (~60 nm Al)**	Deposits and patterns the second layer of **~60 nm sputtered Aluminum** on the wafer surface wherever drawn.

ABSTRACTED DESIGN RULE: DRAW FOR THE FINAL WAFER, NOT THE RAW PHOTOMASK

Do **not** manually invert your polygons for negative photoresist (such as `M1` or `M2` liftoff) or manually combine layers to account for multi-step diffusion/oxidation masks. Draw each layer strictly where you want that physical feature on the wafer—the Photomask Integration Team will generate the actual lithography mask set from your GDSII.

2. WHAT ACTUALLY HAPPENS AT EACH MASK STEP (LAYER BY LAYER)

While you simply draw positive-logic polygons in KLayout, the Photomask Integration Team translates your drawn shapes into the physical lithography masks used in the NRF cleanroom process flow:

- `1. SI_ETCH` **(Mask 1 — Alignment Markers):** Patterns positive photoresist (`AZ1512`) on the starting p-type bulk silicon wafer and uses Reactive Ion Etching (RIE) to etch **~200 nm** into the silicon before the initial ~400 nm wet thermal oxidation (TGO). Most devices will not need to draw on this layer.
- `2. NPLUS` **(Mask 3 — 2nd Annealing / n+ Active Areas) & Why Mask Order Differs:** Physically in the cleanroom, **Mask 2 is technically the** `NWELL` **step** and **Mask 3 is technically the** `NPLUS` **step**. This is because all doped areas must first open the ~400 nm oxide (Mask 2) and undergo the **1st phosphorus Spin-On-Glass (P-SOG) thermal anneal**, which results in moderately doped `n-type` (N-Well) silicon. Next, Mask 3 patterns the wafer so Buffered Oxide Etch (BOE) strips the P-SOG off regions that should remain `NWELL` , while keeping the P-SOG over regions that undergo the **2nd thermal anneal** to become heavily doped **n+ silicon** (`NPLUS`). To make layout much easier for designers, you simply draw `NPLUS` where you want final `n+` silicon and `NWELL` where you want final `n-type` silicon—the Photomask Integration Team automatically combines them for Mask 2 and patterns the P-SOG etch for Mask 3.
- `3. NWELL` **(Mask 2 — 1st Annealing / n Active Areas):** As explained above, this corresponds to the first P-SOG diffusion step. It patterns positive photoresist and uses BOE to etch openings in the initial ~400 nm wet thermally grown oxide (TGO) down to bare silicon, allowing phosphorus from the coated ~200 nm P-SOG film to diffuse into the silicon during the **1st thermal anneal** to form **n-type (N-Well) silicon** (after which the P-SOG in `NWELL` -only areas is stripped before the 2nd anneal).
- `4. THINOX` **(Mask 4 — Thin Dry Oxide):** After all prior TGO and P-SOG are stripped in BOE and a fresh ~400 nm wet TGO is grown across the wafer, this mask step uses BOE to etch away that ~400 nm wet oxide in the drawn regions, followed by dry O_2 thermal oxidation to grow a dense **~40 nm dry oxide** in those regions.
- `5. CONT` **(Mask 5 — Contact Oxide Etch):** Patterns positive photoresist (`AZ1512`) and uses BOE to etch completely through the surface thermal silicon dioxide in the drawn regions, exposing the underlying silicon surface.
- `6. M1` **(Mask 6 — Metal 1 Aluminum Liftoff):** Patterns negative photoresist (`AZ nLOF 2035`) with an undercut profile, sputters **~60 nm of Aluminum** across the wafer using the AJA Sputter tool, and lifts off unwanted metal in `MLO-07` so Aluminum remains only where `M1` was drawn.
- `7. VIA1` **(Mask 7 — PVD SiO_2 Etch):** After Back-End-of-Line (BEOL) begins by sputtering **~1 µm of PVD SiO_2** across the entire wafer, this mask step patterns positive photoresist (`AZ1512`) and uses BOE to etch openings through the ~1 µm PVD SiO_2 layer where drawn.
- `8. M2` **(Mask 8 — Metal 2 Aluminum Liftoff):** Patterns negative photoresist (`AZ nLOF 2035`), sputters the second **~60 nm Aluminum** film via PVD, and performs liftoff in `MLO-07` so the second Aluminum layer remains only where `M2` was drawn.

3. KLAYOUT LAYER REFERENCE & PYTHON GENERATOR TEMPLATE

Below is the reference view of `NRF_Tapeout2_Reference_Example.gds` opened in KLayout, showing the 8 mandatory layers (`1/0` through `8/0`) arranged as individual reference boxes:

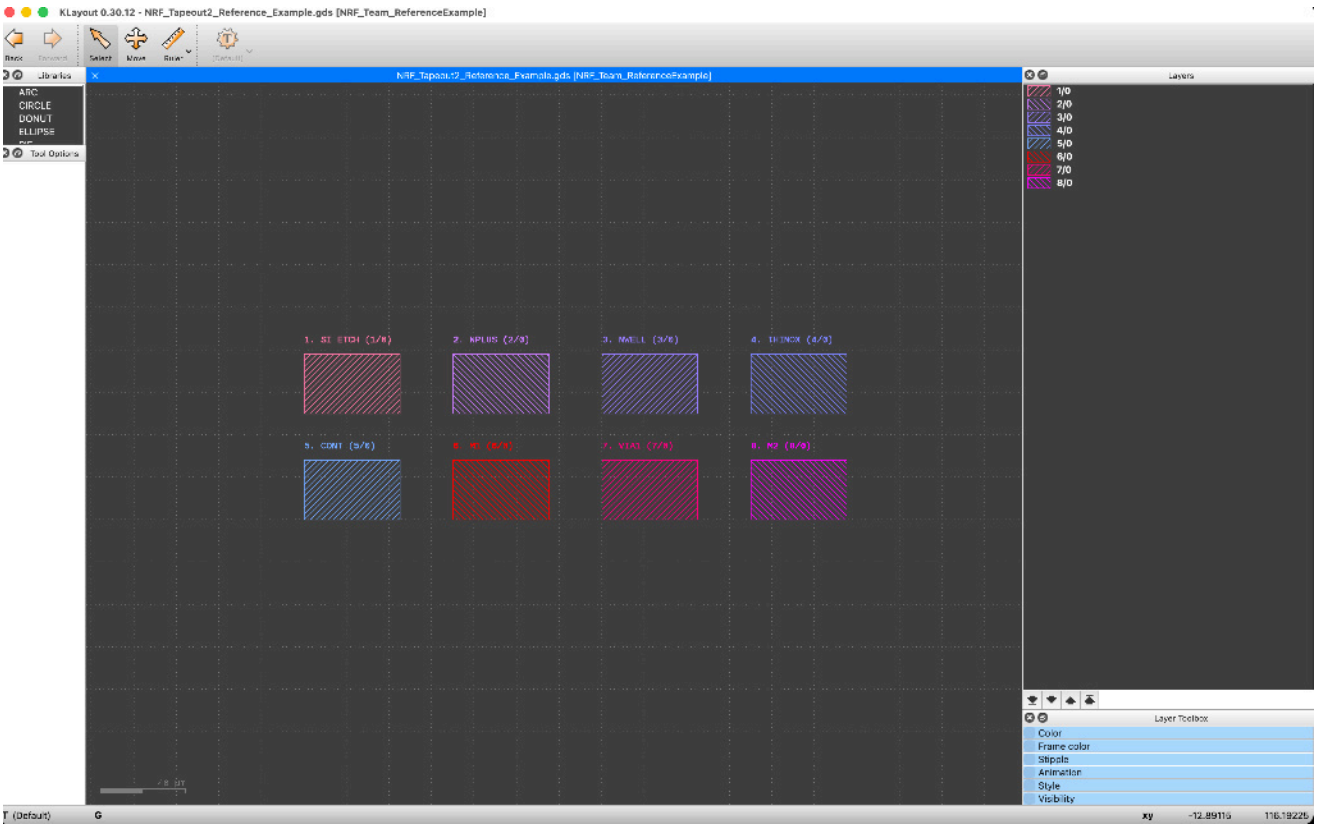


Figure 1: KLayout view of `NRF_Tapeout2_Reference_Example.gds` showing the 8 required layers (`1. SI_ETCH` through `8. M2`).

4. HOW TO USE THE PYTHON SCRIPT TO CREATE YOUR DEVICE GDSII

- **Step 1 — Download the Template Script:** Clone or download the Python generator script from the repository link below.
- **Step 2 — Run the Script to Verify Setup:** Run the Python script (requires `pip install klayout`) or execute it via KLayout's built-in batch interpreter. Running the script automatically generates the `.lyp` layer properties file and the reference GDSII file.
- **Step 3 — Replace the Demo Boxes with Your Device:** Open the Python script, delete the demo block between `# ✂️ DELETE FROM HERE` and `# ✂️ STOP DELETING HERE` , and add your own shapes under `# 🚀 START YOUR DEVICE CODE HERE` . Do not delete the setup block at the top—it pre-registers all 8 layers so even unused layers remain in your exported GDSII file.
- **Step 4 — Save & Inspect in KLayout:** Store your final device GDSII file as `Tapeout2_<DeviceName>.gds` (e.g., `Tapeout2_NMOS_Array.gds`), open it in KLayout, and load the generated `.lyp` file via `File → Load Layer Properties` .

TEMPLATE REPOSITORY & FILES

Repository Link: `https://github.com/ufsilicon/Tapeout2_Layout_Layers_Reference` — Contains the Python template script to generate the KLayout layer properties and your device GDSII file.